



Modelling Financial Markets

Chapter 14: Deep Learning and Time-Series Foundation Models



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Acronyms used in this material (1/2)

- ▣ **ACF**: AutoCorrelation Function
- ▣ **ACI**: Adaptive Conformal Inference
- ▣ **AI**: Artificial Intelligence
- ▣ **AR**: AutoRegressive (model); in event studies: Abnormal Return
- ▣ **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average (model)
- ▣ **BET**: Bucharest Exchange Trading (index)
- ▣ **CI**: Confidence Interval
- ▣ **COVID-19**: COronaVirus Disease 2019
- ▣ **CPU**: Central Processing Unit
- ▣ **CRPS**: Continuous Ranked Probability Score
- ▣ **CRSP**: Center for Research in Security Prices
- ▣ **DM**: Diebold–Mariano (test)
- ▣ **DQ**: Dynamic Quantile (test)
- ▣ **ES**: Expected Shortfall
- ▣ **ETF**: Exchange-Traded Fund
- ▣ **FHS**: Filtered Historical Simulation
- ▣ **FM**: Fama–MacBeth (regressions)
- ▣ **FRED**: Federal Reserve Economic Data
- ▣ **FZ0**: Fissler–Ziegel loss, zero-homogeneous version
- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GIFT**: General Time Series Forecasting Model Evaluation (benchmark GIFT-Eval)

Acronyms used in this material (2/2)

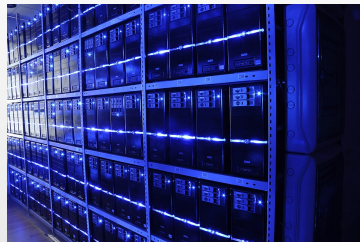
- ▣ **GPD**: Generalised Pareto Distribution
- ▣ **GRU**: Gated Recurrent Unit
- ▣ **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ **HAR**: Heterogeneous AutoRegressive (model)
- ▣ **HS**: Historical Simulation
- ▣ **LASSO**: Least Absolute Shrinkage and Selection Operator
- ▣ **LLM**: Large Language Model
- ▣ **LR**: Likelihood Ratio (test)
- ▣ **LSTM**: Long Short-Term Memory (network)
- ▣ **MCS**: Model Confidence Set
- ▣ **MDE**: Minimum Detectable Effect
- ▣ **MSE**: Mean Squared Error
- ▣ **NBER**: National Bureau of Economic Research
- ▣ **PIT**: Probability Integral Transform
- ▣ **QLIKE**: Quasi-LIKElihood (loss function)
- ▣ **QS**: Quantile Score
- ▣ **RFF**: Random Fourier Features
- ▣ **RNN**: Recurrent Neural Network
- ▣ **RV**: Realised Variance
- ▣ **SPA**: Superior Predictive Ability (test)
- ▣ **TS**: Time Series (in the name TS-Arena)
- ▣ **US**: United States
- ▣ **VAR**: Vector AutoRegression

Chapter Route and Outcomes

- Question: do deep networks and time-series **foundation models** forecast **returns**, **volatility** and **tail risk** better than the classical models of [Chapters 5–9](#)?
 - ▶ **Deep learning**: models built from many layers of simple non-linear units, trained on data; since 2024, foundation models pretrained on billions of observations
 - ▶ Answered with the evaluation tools of [Chapter 8](#): strong benchmarks, loss differentials, backtests
- Route of the chapter
 - ▶ Part I: networks for sequences, attention, foundation models, Test 1 (returns)
 - ▶ Part II: Test 2 (realised volatility), Test 3 (VaR 1% and ES 2.5%), case study, AI for scientific discovery
- By the end you can
 - ▶ explain vanishing gradients as VAR impulse responses and the LSTM cell state as an AR(1)
 - ▶ turn a zero-shot quantile grid into point, volatility and VaR/ES forecasts
 - ▶ test a new model correctly: nesting, multiplicity, calibration, contamination and power

Three Forecasting Targets

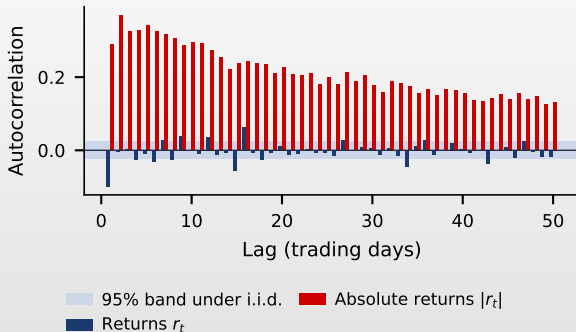
- ▣ **Returns** r_t (conditional mean)
 - ▶ Very low signal-to-noise ratio; predictable patterns are traded away ([Chapter 2](#))
 - ▶ Benchmark: the zero forecast
- ▣ **Volatility** (conditional variance)
 - ▶ Persistent and clustered ([Chapters 1 and 5](#)): there is a signal to learn
 - ▶ Benchmark: log-HAR
- ▣ **Tail-risk measures**: VaR 1% and ES 2.5%
 - ▶ Need the shape of the lower tail, not just the centre of the distribution ([Chapter 7](#))
 - ▶ Benchmark: FHS



A data centre: deep networks are trained on racks of such servers

Photo: [BalticServers.com](#) (2013); CC BY-SA 3.0; [Wikimedia Commons](#)

Where Is the Signal?



- S&P 500 daily returns since 2000 (6 718 days), lags 1–50
 - ▶ Light-blue band: $\pm 1.96/\sqrt{T} = \pm 0.024$, the 95% range of an autocorrelation under independence
 - ▶ T : number of days
- Returns: lag-1 autocorrelation -0.099
 - ▶ 15 of 50 lags outside the band, small and of changing sign
 - ▶ Little to learn about the mean
- Absolute returns: 0.29 at lag 1, 0.13 at lag 50
 - ▶ All 50 lags outside the band: persistent volatility clustering
- Prediction for the chapter
 - ▶ Flexible models can find the volatility signal, not much in the mean

 [MFM_ch14_acf_signal](#)

From the Multilayer Perceptron to Deep Learning (1/2)

- MLP (multilayer perceptron, [Chapter 13](#)): each layer applies a non-linear function to a linear combination of the previous layer

$$h^{(\ell)} = \varphi \left(W^{(\ell)} h^{(\ell-1)} + b^{(\ell)} \right), \quad h^{(0)} = x$$

- ▶ x : input vector; $h^{(\ell)}$: output of layer ℓ ; $W^{(\ell)}$, $b^{(\ell)}$: weight matrix and bias vector of layer ℓ , estimated from the data
- ▶ φ : activation, applied element by element: ReLU $\max(0, z)$, tanh, or the sigmoid $\sigma(z) = 1/(1 + e^{-z}) \in (0, 1)$
- Training: backpropagation ([Rumelhart, Hinton & Williams \(1986\)](#)) computes the gradient of the loss with respect to every weight; Adam ([Kingma & Ba \(2015\)](#)) is the update rule
- Losses: mean squared error (MSE) for point forecasts; for a quantile of level u , the pinball loss

$$\rho_u(y, q) = (u - 1\{y < q\})(y - q)$$

- ▶ y : realised value; q : quantile forecast; $1\{\cdot\}$: 1 if the condition holds, 0 otherwise
- ▶ asymmetric: values above q cost u per unit, values below cost $1 - u$; the expected loss is minimised by the true u -quantile

From the Multilayer Perceptron to Deep Learning (2/2)

- Universal approximation ([Hornik, Stinchcombe & White \(1989\)](#))
 - ▶ A wide enough network approximates any continuous function
 - ▶ Silent on the sample size needed, the binding constraint with one financial series
 - ▶ Regularisation: early stopping, weight decay (a penalty on the size of the weights), ensembles over seeds; the validation block must follow the training block
- 2024 Nobel Prize in Physics to Hopfield and Hinton
 - ▶ The methods: [LeCun, Bengio & Hinton \(2015\)](#)
 - ▶ In finance, [Gu, Kelly & Xiu \(2020\)](#) gain on a cross-section of thousands of stocks, not on one index



Geoffrey Hinton, Nobel laureate 2024

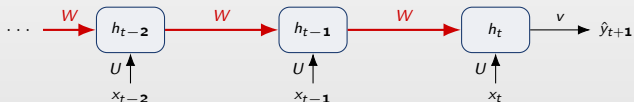
Photo: [Arthur Petron \(2024\)](#); CC BY-SA 4.0; [Wikimedia Commons](#)

RNN and LSTM in One Slide (1/2)

- **Recurrent neural network (RNN):** a hidden state is updated at every step from its previous value and the new input

$$h_t = \tanh(W h_{t-1} + U x_t + b), \quad \hat{y}_{t+1} = v^\top h_t + c$$

- ▶ h_t : hidden state, the memory of the network at t ; x_t : input at t (e.g. today's return); $\hat{y}_{t+1}$: forecast
- ▶ W , U , b : recurrent weights, input weights and bias; v , c : output weights and intercept; $\tanh$ keeps h_t in $(-1, 1)$
- ▶ The same weights at every step: unrolled over time, a very deep network with shared weights



RNN and LSTM in One Slide (2/2)

- **Long short-term memory** (LSTM, Hochreiter & Schmidhuber (1997)): gates decide what to forget, what to write and what to show

$$\begin{aligned}f_t, i_t, o_t &= \sigma(W_{\bullet}x_t + U_{\bullet}h_{t-1} + b_{\bullet}), & g_t &= \tanh(W_gx_t + U_g h_{t-1} + b_g) \\c_t &= f_t \odot c_{t-1} + i_t \odot g_t, & h_t &= o_t \odot \tanh(c_t)\end{aligned}$$

- Notation

- ▶ f_t : forget gate, i_t : input gate, o_t : output gate; each has its own weights $W_{\bullet}$, $U_{\bullet}$, $b_{\bullet}$, $\bullet \in \{f, i, o\}$
- ▶ σ : sigmoid, values in $(0, 1)$: a gate near 0 blocks, near 1 lets through
- ▶ g_t : candidate new content; c_t : cell state, the long-term memory; h_t : output; $\odot$: element-by-element product

- GRU (gated recurrent unit): a lighter variant with two gates

Vanishing Gradients: a Bound and a VAR(1) (1/2)

- Training needs the effect of old states on today's state; backpropagation through time multiplies one Jacobian per step

$$\frac{\partial h_T}{\partial h_{T-k}} = \prod_{j=0}^{k-1} D_{T-j} W, \quad D_t = \text{diag}(\varphi'(Wh_{t-1} + Ux_t + b))$$

- ▶ $\partial h_T / \partial h_{T-k}$: the matrix of changes in h_T per unit change in the state k steps earlier
- ▶ D_t : diagonal matrix of the slopes φ' of the activation at step t

- Bound, by submultiplicativity $\|AB\|_2 \leq \|A\|_2 \|B\|_2$ (Pascanu, Mikolov & Bengio (2013))

$$\left\| \frac{\partial h_T}{\partial h_{T-k}} \right\|_2 \leq (\gamma_\varphi \|W\|_2)^k, \quad \gamma_\varphi = \sup_z |\varphi'(z)|$$

- ▶ $\|W\|_2$: spectral norm, the largest factor by which W stretches a vector; γ_φ : the largest slope of the activation (= 1 for tanh)
- ▶ $\gamma_\varphi \|W\|_2 < 1$: the gradient shrinks geometrically with k , i.e. it vanishes

- Scalar case $|w|^k$: $0.9^{20} = 0.12$, $0.9^{50} = 0.005$ (Bengio, Simard & Frasconi (1994)); remedies: clipping the gradient norm (explosion), gating as in the LSTM (vanishing)

Vanishing Gradients: a Bound and a VAR(1) (2/2)

- Linearise around a fixed point h^* of the network (a state that reproduces itself when the input is zero)

$$h_t - h^* \approx A(h_{t-1} - h^*) + DUx_t, \quad A = DW$$

- ▶ D : the slopes of the activation at h^* ; the hidden state follows a VAR(1) (vector autoregression) with coefficient matrix A
- $\partial h_T / \partial h_{T-k} = A^k$ is the impulse response of this VAR at horizon k
 - ▶ $\rho(A)$: spectral radius, the largest absolute eigenvalue of A
 - ▶ $\rho(A) < 1$: stable VAR, gradients decay like $\rho(A)^k$; $\rho(A) > 1$: they explode
- A stable RNN near its fixed point is a short-memory model, like any stationary VAR (Seminar A1; details in the [Appendix](#))

Why the LSTM Remembers

- Along the cell state the only multiplier is the forget gate
 - ▶ $\partial c_T / \partial c_{T-k} = \prod_{j=0}^{k-1} f_{T-j}$ on the direct path
 - ▶ $c_t = f_t \odot c_{t-1} + i_t \odot g_t$: an AR(1) with a random coefficient
- Constant forget gate $f = \sigma(b_f)$, b_f its bias: an AR(1) with coefficient f
 - ▶ $b_f = 0$: $f = 0.5$, $0.5^{50} \approx 9 \times 10^{-16}$
 - ▶ $b_f = 3$: $f = 0.953$, $0.953^{50} \approx 0.09$; half-life 14 steps (the k with $f^k = \frac{1}{2}$, $k = \ln 0.5 / \ln f$)
- Short memory by construction: $\text{Corr}(c_t, c_{t-k}) = f^k$ decays geometrically
 - ▶ Long memory decays hyperbolically: $\rho_k \sim Ck^{2d-1}$, with ρ_k the autocorrelation at lag k , $d \in (0, \frac{1}{2})$ the memory parameter, $C > 0$ a constant
 - ▶ Fitted to the S&P 500 absolute returns, $f = 0.984$ predicts 0.027 at lag 150 against 0.059 observed ([▶ Appendix](#))

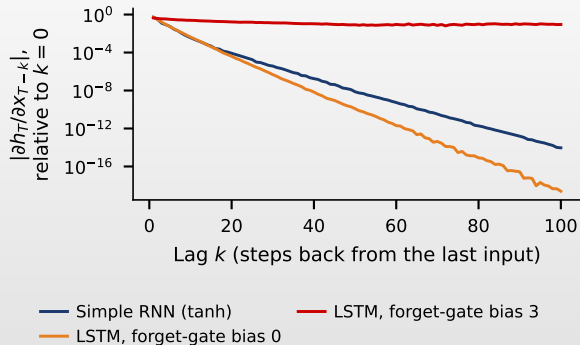


Sepp Hochreiter, co-inventor of the LSTM

Photo: Eulenreich (2015); CC BY-SA 4.0;

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The Vanishing Gradient, Measured



- $|\partial h_T / \partial x_{T-k}|$ by automatic differentiation
 - ▶ 16 hidden units, 50 random initialisations, same Normal inputs
- Simple RNN: 8.3×10^{-5} at lag 20, 7.4×10^{-9} at lag 50
 - ▶ Nothing is learned from inputs more than a few weeks back
- LSTM, forget-gate bias 0: 1.2×10^{-10} at lag 50
 - ▶ No better than the RNN: $f \approx 0.5$ halves the memory at every step
- LSTM, bias 3: 8.4% at lag 50
 - ▶ As predicted by $0.953^{50} \approx 0.09$
 - ▶ Initialise the forget-gate bias at a positive value

Attention, Positions and Patches (1/2)

- Each time step asks a question (query) and compares it with every past step (key); the answer is a weighted average of their values (Vaswani et al. (2017))

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \quad Q = XW_Q, \quad K = XW_K, \quad V = XW_V$$

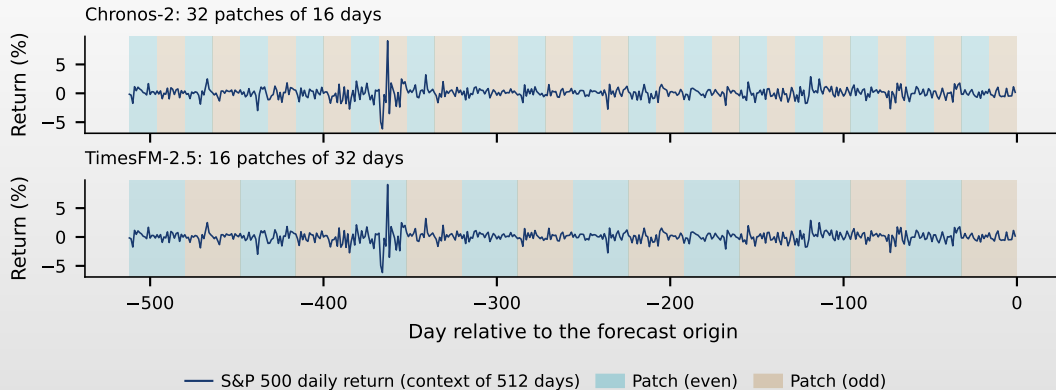
- Notation

- ▶ X : the inputs, one row per time step; W_Q, W_K, W_V : learned weight matrices; Q, K, V : queries, keys and values
 - ▶ $QK^\top$: the similarity score of every pair of steps; d_k : length of a key vector; dividing by $\sqrt{d_k}$ keeps the scores from saturating the softmax
 - ▶ softmax: turns each row of scores s_j into weights $e^{s_j} / \sum_i e^{s_i}$ that sum to one
- No product of Jacobians over time: any past step is one hop away; cost $O(n^2)$: the work grows with the square of the number of inputs n

Attention, Positions and Patches (2/2)

- **Positional encoding:** attention treats its inputs as a set, so order is added (sinusoids in the original Transformer; rotary embeddings in Chronos-2)
- **Patching** (Nie et al. (2023)): a token is a block of consecutive observations
 - ▶ 512 daily returns in patches of 16 days: 32 tokens, about 256 times fewer attention scores (32^2 instead of 512^2); each token summarises a few weeks of market activity

Patching a 512-Day Context (1/2)



Patching a 512-Day Context (2/2)

- The last 512 daily returns of the S&P 500, split as Chronos-2 (16 days) and TimesFM-2.5 (32 days) read them
- Each shaded block becomes one embedding vector for the Transformer
- Global models learn across many series
 - ▶ DeepAR ([Salinas et al. \(2020\)](#)), N-BEATS ([Oreshkin et al. \(2020\)](#)), PatchTST ([Nie et al. \(2023\)](#))
 - ▶ Foundation models push this idea to very large corpora

 MFM_ch14_tokenization

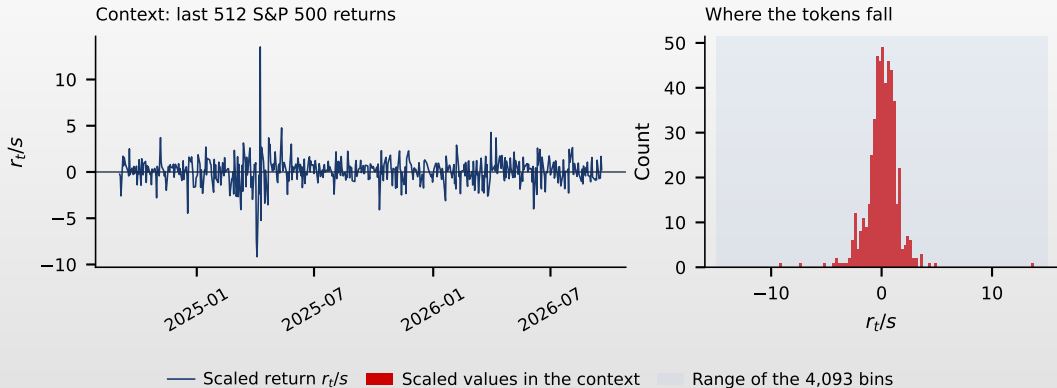
What Is a Foundation Model?

- A large network pretrained once on a very broad corpus, then used on new tasks
 - ▶ Idea borrowed from large language models (LLM): pretrain on text, then use on any question
- **Zero-shot** forecasting: no training on the target series; only its recent history (the **context**) is passed at forecast time
 - ▶ Output: a set of quantiles of the future values, not only a point
 - ▶ Fine-tuning (further training on the target series) is possible but not used here
- Appeal for finance: one model for thousands of series, no estimation per series

Chronos: Time Series as a Language

- ▣ Ansari et al. (2024): turn values into tokens and train a language model (T5) with cross-entropy
 - ▶ Mean scaling: divide the context by $s = \frac{1}{C} \sum_t |x_t|$, the mean absolute value of the C values x_t in the context (its length)
 - ▶ Quantisation: 4,093 equally spaced bin centres in $[-15, 15]$ (width 0.0073), plus special tokens
 - ▶ Forecast: sample future tokens, map back to values, read quantiles from the samples
- ▣ Pretraining corpus: public datasets (energy, transport, retail, weather, web, finance and economics) plus synthetic series (TSMixup, KernelSynth)
 - ▶ Model sizes 20 to 710 million parameters; the small version has 46 million

What Chronos Sees: Scaled Returns and Tokens (1/2)



What Chronos Sees: Scaled Returns and Tokens (2/2)

- Context: the last 512 S&P 500 daily returns; scale $s = 0.672\%$; scaled returns from -9.2 to 13.5 , 96% within ± 3
- The 512 returns use 336 distinct tokens out of 4,093: scaling makes the model free of units, so the volatility level must be read from the context
- A crash day with scaled return M raises s by about $|M|/C$ (C : context length): strongly in a short context, little in a 512-day one (Seminar, A3)

 MFM_ch14_tokenization

The Three Models We Test

Model	Parameters	Patch	Quantile levels	Weights released
Chronos-Bolt (small)	48 M	16	10%–90%	25.11.2024
TimesFM-2.5	231 M	32	mean, 10%–90%	01.10.2025
Chronos-2	119 M	16	1%, 5%, ..., 99%	30.10.2025

- **Chronos-Bolt**: patches and a direct quantile output instead of token sampling
- **Chronos-2** ([Ansari et al. \(2025\)](#)): robust scaling, attention across time and across groups of related series
- **TimesFM** ([Das et al. \(2024\)](#)): decoder-only Transformer, pretrained on about 10^{11} time points
- All run on a CPU with a 512-day context; the release dates of the weights bound what the models could have seen; only Chronos-2 reaches the 1% level

From a Quantile Grid to VaR and ES (1/2)

- The model returns quantiles q_u at levels $u_1 < \dots < u_K$: $P(Y \leq q_u) = u$, with Y the future value and K the number of levels
- Point forecasts from the grid
 - ▶ the median $q_{0.5}$, or the mean $\int_0^1 q_u du$, the average of all quantiles
 - ▶ for an increasing function g : $E[g(Y)] = \int_0^1 g(q_u) du$ (e.g. $g = \exp$ when Y is a log variance)
- Tail risk at level α (Chapter 7), for a return Y

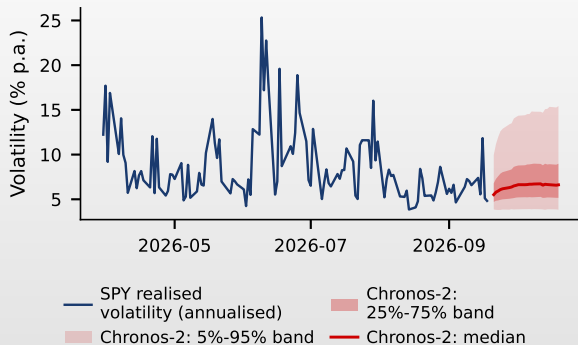
$$\text{VaR}_\alpha = -q_\alpha, \quad \text{ES}_\alpha = -\frac{1}{\alpha} \int_0^\alpha q_u du$$

- ▶ VaR: the loss exceeded with probability α ; ES: the average loss on the worst α share of days
- Below the lowest level the grid is silent: holding q_u at $q_{0.01}$ understates ES, interpolation can err either way (Seminar, A5–A6)
 - ▶ Chronos-Bolt and TimesFM stop at 10%: a "VaR 1%" request returns the 10% quantile

From a Quantile Grid to VaR and ES (2/2)

- Worked example, S&P 500, day after 18.09.2026: $q_{0.01} = -2.70\%$, $q_{0.05} = -1.40\%$
 - ▶ VaR 1% = 2.70%
 - ▶ linear interpolation between the two levels: $q_{0.025} = q_{0.01} + \frac{0.015}{0.04}(q_{0.05} - q_{0.01}) = -2.21\%$, so VaR 2.5% = 2.21%
 - ▶ ES 2.5%: the integral split at 1%, q_u held at $q_{0.01}$ below 1% and averaged by the trapezoid rule above
- $ES_{2.5\%} = -\frac{1}{0.025} \left[0.01 q_{0.01} + 0.015 \frac{q_{0.01} + q_{0.025}}{2} \right] = 2.56\%$
- Reading: ES 2.5% exceeds VaR 2.5%, as it always does, because it averages the losses beyond VaR; here it is close to VaR 1%

A Zero-Shot Fan Chart for Volatility



- Chronos-2 on log realised variance of SPY (512-day context)
 - ▶ 22 trading days ahead from 18.09.2026
 - ▶ Bands: quantiles 5–95% and 25–75%
- Last volatility 4.8% p.a., 20-day average 6.6%
 - ▶ Median forecast at 22 days 6.6%, 90% band 3.8–15.4%
- The band widens and is skewed upwards: mean reversion, as in GARCH and HAR ([Chapters 5, 9](#)), with no estimation on SPY
- Whether these forecasts are better is the question of Tests 1–3

 MFM_ch14_fan_chart

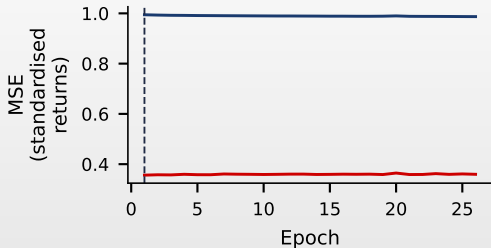
Pretraining Leakage and Contamination

- A test is out of sample only if the test period was never seen during training; four ways it can be “seen”
 - ▶ (i) the same series is in the pretraining corpus; (ii) correlated series carry the same shocks; (iii) developers select model versions on overlapping public benchmarks
 - ▶ (iv) our own look-ahead: scalings, windows, seeds or contexts chosen with test data
- The Chronos corpus includes financial series (e.g. daily exchange rates 1990–2016); we cannot rule out that our series were seen
- Remedy for (i)–(iii): a **post-release window** from 3 November 2025, a common cutoff after all three sets of weights were published
 - ▶ Clean but short (about 220 trading days; 320 calendar days for Bitcoin): low power, especially for VaR 1%
- (iv) is under our control: every scaling, window and estimate uses only past data; a formal protocol is in the [Appendix](#)

Design of the Return Test

- Daily log returns (%): S&P 500 and BET from 04.01.2016 (2 693 and 2 675 days), Bitcoin from 01.01.2019 (2 818 days)
 - ▶ Every forecast uses only data up to the previous close
- Models: historical mean, AR(1) on a 1,000-day window, LSTM, Chronos-2, Chronos-Bolt and TimesFM-2.5 (median)
 - ▶ Squared error targets the mean; grid means ($\int_0^1 q_u du$) do worse still, e.g. S&P 500 R^2 -0.77% (Chronos-2), -3.42% (TimesFM-2.5)
- Benchmark: the zero forecast; out-of-sample $R^2 = 1 - \sum_t (r_t - \hat{r}_t)^2 / \sum_t r_t^2$
 - ▶ r_t : realised return; $\hat{r}_t$: forecast from data up to $t - 1$; $R^2 > 0$: smaller squared errors than the zero forecast
 - ▶ Zero rather than the historical mean, as in [Gu, Kelly & Xiu \(2020\)](#); compare [Campbell & Thompson \(2008\)](#)
 - ▶ Inference: block-bootstrap CI for R^2 (blocks of 20 days), Diebold–Mariano (DM) test on squared errors with Newey–West variance ([Diebold & Mariano \(1995\)](#))

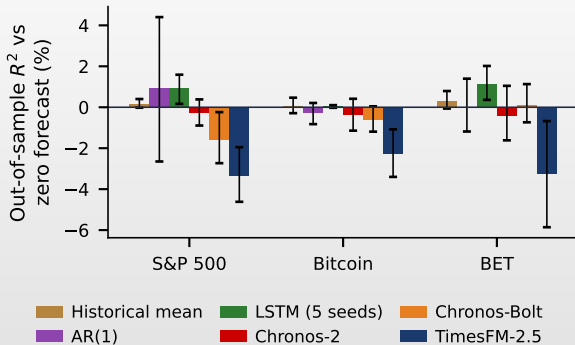
The LSTM for Returns and Early Stopping



- Training MSE
- Validation MSE (last 20% of the training period)
- - - Selected epoch (early stopping)

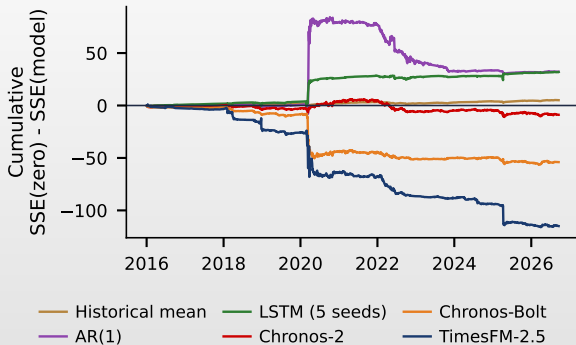
- Input: the last 20 returns, standardised on the training period
 - ▶ $h = 16$ hidden units, $d = 1$ input per step, linear output
 - ▶ $4(hd + h^2 + 2h) + h + 1 = 1\,233$ parameters; 4: the three gates and the candidate
- MSE, Adam, weight decay 10^{-4}
 - ▶ Trained once on up to 4,000 days, the last 20% for early stopping
 - ▶ Forecast = average of 5 seeds
- Chart (S&P 500, seed 0): training stops 26 epochs in, the weights of epoch 1 are kept
- Little to learn: forecast s.d. 0.047% against 1.13% for returns
 - ▶ Validation MSE is lower only because those years were calmer

Out-of-Sample R^2 for Returns



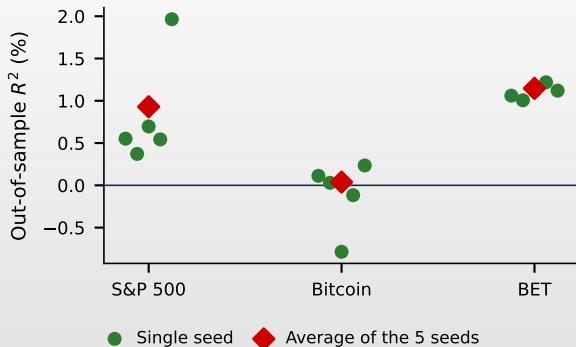
- R^2 against the zero forecast (%), 95% block-bootstrap CIs
 - ▶ Full table with DM p -values in the [Appendix](#)
- S&P 500 and Bitcoin: no model beats zero significantly
- TimesFM-2.5 is significantly **worse** than zero
 - ▶ S&P 500: R^2 -3.35% , DM $p = 0.001$
 - ▶ Forecast s.d. 0.141% , three times the LSTM's
- BET: LSTM R^2 $+1.15\%$ (DM $p = 0.014$)
 - ▶ A small signal, tested below for nesting, multiplicity and stale prices

Cumulative Squared-Error Gains on the S&P 500



- Curve: $\sum_{s \leq t} [r_s^2 - (r_s - \hat{r}_s)^2]$
 - Rises when the model beats the zero forecast, falls when it loses
- All curves stay near zero in calm years and jump in March 2020
 - One month decides the ranking
- AR(1) gains in 2020 from the negative autocorrelation of crash days
 - Then gives part of the gain back; TimesFM-2.5 loses steadily
- Report the path, not only the final R^2

The Seed Lottery



- Out-of-sample R^2 of five LSTMs that differ only in the random seed, and of their average
- Bitcoin: single seeds from -0.79% to $+0.24\%$; S&P 500 from $+0.37\%$ to $+1.96\%$: variation as large as the effect
- BET: all 5 seeds between $+1.01\%$ and $+1.22\%$: robust across seeds, not yet robust to multiplicity or stale prices
- Choosing the best seed after seeing the test is backtest overfitting ([Chapter 13](#))

Nested Models and Many Tests (1/2)

- Zero is nested in the historical mean and AR(1): under H_0 the larger model only adds estimation noise, so DM rejects too rarely
- Clark–West (Clark & West (2007)): add back the noise term and test its mean, one-sided HAC t -test

$$\tilde{d}_t = e_{0,t}^2 - \left[e_{1,t}^2 - (\hat{r}_{0,t} - \hat{r}_{1,t})^2 \right]$$

- ▶ $\hat{r}_{0,t}$: zero forecast (small model); $\hat{r}_{1,t}$: forecast of the large model; $e_{0,t}$, $e_{1,t}$: their errors $r_t - \hat{r}_{\cdot,t}$
 - ▶ $(\hat{r}_{0,t} - \hat{r}_{1,t})^2$: the cost of estimating coefficients that are zero under H_0 ; HAC: heteroskedasticity and autocorrelation consistent
 - ▶ H_0 : $E[\tilde{d}_t] = 0$; a large positive t : the large model adds information
- One-sided p , DM vs Clark–West: historical mean 0.048 vs 0.007 (S&P 500), 0.041 vs 0.003 (BET); AR(1) 0.320 vs 0.055

Nested Models and Many Tests (2/2)

- ▣ 6 models $\times$ 3 markets = 18 tests; the smallest one-sided DM p is the BET LSTM, 0.007
 - ▶ Bonferroni: reject only $p < 0.05/M$ for M tests, here 0.0028; Holm: the same, applied step by step from the smallest p ; both control the probability of any false rejection
 - ▶ Holm-adjusted p over the 18 tests 0.130: not significant
- ▣ Within the six BET models, tests that use the correlation between the models
 - ▶ Romano–Wolf ([Romano & Wolf \(2005\)](#)): step-down adjusted p 0.016
 - ▶ SPA ([Hansen \(2005\)](#)), H_0 : no model beats zero: BET 0.018, S&P 500 0.130, Bitcoin 0.354
- ▣ Reading: “some model beats zero on the BET” survives; “one of 18 comparisons is real” does not. Fix the family before looking

Is BET Predictability Real? Non-Synchronous Trading

- Stale prices of thinly traded constituents make index returns positively autocorrelated, even if every price is efficient (Lo & MacKinlay (1990))
 - ▶ First-order autocorrelation since 2016: BET 0.06, S&P 500 -0.14
 - ▶ The LSTM forecast has correlation 0.88 with the previous day's BET return: it mostly learned this autocorrelation
- Tradable check: the BET-tracking ETF Patria-TVBETETF, 2629 common days, same LSTM signal
 - ▶ Same day: R^2 2.88% (BET 1.26%), but the ETF is thinly traded too (autocorrelation 0.07, 4.2% zero-return days)
 - ▶ Executed one day later: R^2 0.57%, Clark–West $p = 0.003$
- Reading: the evidence is consistent with stale-price effects; a small signal survives the execution lag, before transaction costs (Chapter 2)

Target, Benchmark and Loss (1/2)

- Realised variance (RV, Andersen et al. (2003)): sum of squared 5-minute log returns within the trading day

$$RV_t = \sum_{j=1}^{78} r_{t,j}^2$$

- $r_{t,j}$: log return of the j -th 5-minute interval of day t ; 78 intervals in the regular SPY session (6.5 hours); average volatility 11.9% p.a.

- HAR (Corsi (2009)): today's RV regressed on yesterday's, the last week's and the last month's averages

$$RV_t = \beta_0 + \beta_d RV_{t-1} + \beta_w \overline{RV}_{t-5:t-1} + \beta_m \overline{RV}_{t-22:t-1} + \varepsilon_t$$

- $\overline{RV}_{t-5:t-1}$, $\overline{RV}_{t-22:t-1}$: averages over the last 5 and 22 trading days; β_d , β_w , β_m : daily, weekly and monthly weights; ε_t : error
 - three averages of different length mimic long memory with few parameters

Target, Benchmark and Loss (2/2)

- log-HAR: the same regression on log RV; forecast $\exp(\hat{y} + \hat{s}^2/2)$
 - ▶ $\hat{y}$: fitted log RV; $\hat{s}^2$: variance of the log residuals; the factor $e^{\hat{s}^2/2}$ turns a log forecast into the mean of RV if the log error is conditionally Normal
- **QLIKE** loss (Patton (2011)) of a variance forecast $\hat{h}_t$

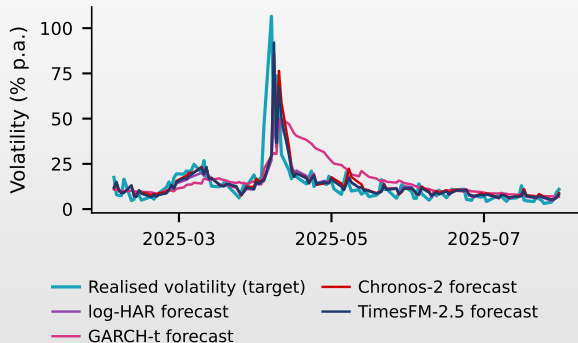
$$L = \frac{RV_t}{\hat{h}_t} - \log \frac{RV_t}{\hat{h}_t} - 1$$

- ▶ $L \geq 0$, and $L = 0$ when $\hat{h}_t = RV_t$; under-forecasts cost more than over-forecasts of the same size
- ▶ Minimised by the conditional mean; ranks forecasts as the true variance σ_t^2 would when RV is a conditionally unbiased proxy, $E[RV_t | \mathcal{F}_{t-1}] = \sigma_t^2$, with $\mathcal{F}_{t-1}$ the information up to $t - 1$ (so does MSE)

Design of the Volatility Test

- ▣ SPY realised variance, one-day-ahead forecasts from 11.10.2022 to 18.09.2026 (979 days)
 - ▶ HAR and log-HAR on a rolling 500-day window; random walk: $\hat{h}_t = \text{RV}_{t-1}$
 - ▶ GARCH-t on daily SPY returns, rescaled to RV units by the ratio of average RV to average squared return over 500 days
 - ▶ LSTM on the last 22 values of log RV, retrained every 125 days on 500 days (hidden size 8, three seeds)
- ▣ Foundation models on log RV (context 512): forecast $\hat{h}_t = \int_0^1 \exp(q_u) du$ on the quantile grid: q_u are quantiles of log RV, $\exp(q_u)$ those of RV, and their average is the mean of RV
 - ▶ Post-release window: 218 days from 3 November 2025

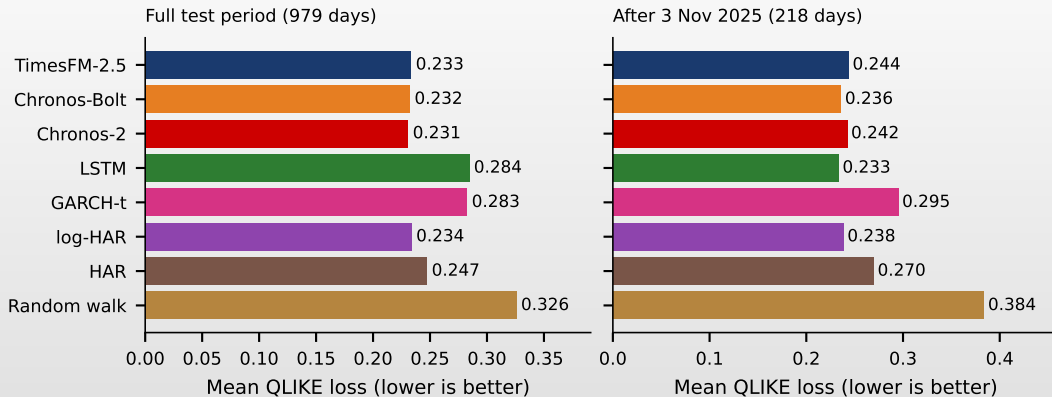
Forecasts Through the April 2025 Shock



- SPY realised volatility (annualised) and one-day-ahead forecasts, February–July 2025
 - ▶ Peak 107% p.a. on 07.04.2025, after the US tariff announcements
- No model anticipates the jump; all react the next day: one-day forecasts extrapolate the past, shocks come from news
- Foundation models and log-HAR decay back at similar speeds; GARCH-t reacts to daily returns and decays more slowly

MFM_ch14_realised_vol

QLIKE Losses of Eight Volatility Models (1/2)



MFM_ch14_realised_vol

QLIKE Losses of Eight Volatility Models (2/2)

- Left: full test period; right: after 3 November 2025 (218 days); full table with DM and MCS p -values in the [Appendix](#)
- Chronos-2 (0.231), Chronos-Bolt, TimesFM-2.5 and log-HAR (0.234): DM tests not significant (Chronos-2 $p = 0.537$), all in the 90% MCS
- Random walk and GARCH-t are eliminated; HAR in levels over-forecasts (mean forecast / mean RV = 1.36); after release the small LSTM is best (0.233): rankings are unstable
- No conditional difference: Giacomini–White test against log-HAR, with instruments 1 and yesterday's loss differential d_{t-1} , $p = 0.760$ (Chronos-2), 0.935 (TimesFM-2.5); no difference detected is not proof of equality

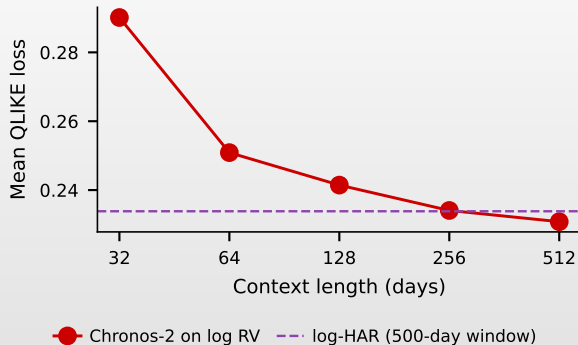
 MFM_ch14_realised_vol

Median or Mean? A Costly Detail

Chronos-2 and TimesFM-2.5 on log RV	QLIKE	Mean forecast / mean RV
Chronos-2, $\int_0^1 \exp(q_u) du$	0.231	1.02
Chronos-2, $\exp(q_{0.5})$	0.259	0.79
TimesFM-2.5, $\int_0^1 \exp(q_u) du$	0.233	0.94
TimesFM-2.5, $\exp(q_{0.5})$	0.259	0.79

- A theorem, not a curiosity ([Gneiting \(2011\)](#)): QLIKE is consistent for the mean: the forecast h that minimises its expected value given the information $\mathcal{F}$ is the conditional mean, $\arg \min_h E[L(RV, h) | \mathcal{F}] = E[RV | \mathcal{F}]$
 - ▶ Monotone maps preserve quantiles: $\exp(\text{median of log RV})$ is the median of RV, elicited by absolute error, not by QLIKE
 - ▶ Lognormal RV, s^2 the variance of log RV: median/mean = $e^{-s^2/2}$; the median version is significantly worse (DM +3.60 for Chronos-2) and worse than log-HAR (0.234)

How Much Context Does Chronos-2 Need?



- QLIKE of Chronos-2 on log RV with contexts of 32 to 512 days, on the same forecast days
- QLIKE falls from 0.290 (32 days) to 0.231 (512 days); below about 256 days Chronos-2 is worse than log-HAR (0.234)
- Long memory: months of history reveal the slow-moving level, which HAR encodes explicitly through its 22-day average
- Context length is a modelling choice to report and test, like a window length

Design of the Tail-Risk Test and the FZ0 Loss (1/2)

- Targets: $\text{VaR}_\alpha = -q_\alpha(r_{t+1} \mid \mathcal{F}_t)$, the loss exceeded with probability α , and $\text{ES}_\alpha = -\frac{1}{\alpha} \int_0^\alpha q_u du$, the mean loss beyond it
 - ▶ $q_\alpha(r_{t+1} \mid \mathcal{F}_t)$: the α -quantile of tomorrow's return given today's information $\mathcal{F}_t$
 - ▶ levels: VaR 1% (coverage tests) and the pair (VaR 2.5%, ES 2.5%)
- Classical models, 1,000-day window, re-estimated every 20 days: historical simulation (HS), GARCH(1,1)-t, filtered historical simulation (FHS, [McNeil & Frey \(2000\)](#))
 - ▶ **Raw**: Chronos-2 quantiles of tomorrow's return (context: 512 returns)
 - ▶ **Hybrid**: scale $\hat{\sigma}_t = \text{forecast of } |r_t|$ (Chronos-2: grid mean; TimesFM: median); FHS on $r_t/\hat{\sigma}_t$: the empirical quantile of the standardised returns, times $\hat{\sigma}_t$

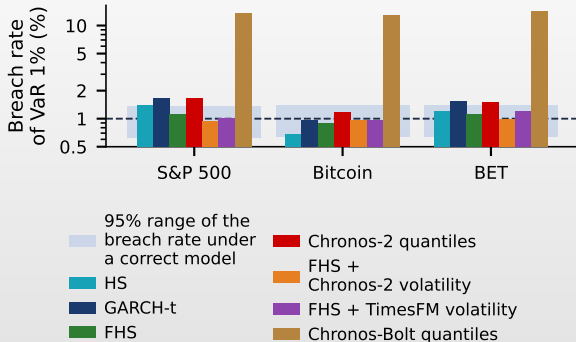
Design of the Tail-Risk Test and the FZ0 Loss (2/2)

- A functional is elicitable if some loss is minimised, in expectation, by its true value; ES alone is not, the pair (VaR, ES) is (Fissler & Ziegel (2016))
- FZ0 loss (Patton, Ziegel & Chen (2019)) of a forecast pair, with the daily loss $L_t = -r_t$

$$\text{FZ0} = \frac{1\{L_t > v\}(L_t - v)}{\alpha e} + \frac{v}{e} + \log e - 1$$

- ▶ v : VaR forecast; $e > 0$: ES forecast; α : tail level; $1\{L_t > v\}$: 1 on a breach day
- ▶ Lower is better: the first term penalises breaches, $\log e$ penalises wide forecasts
- Models compared by DM tests on FZ0 differences and by the model confidence set (MCS, Hansen, Lunde & Nason (2011)), the set of models not significantly worse than the best

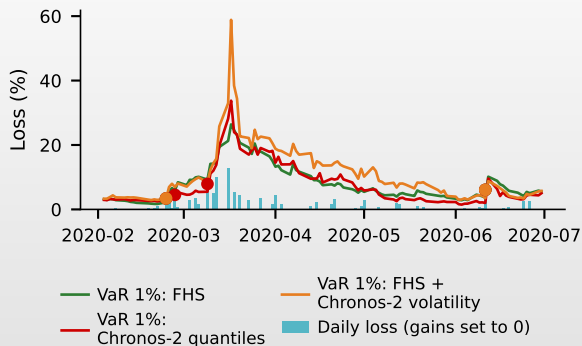
VaR 1% Breach Rates



- Share of days with a loss above VaR 1% (log scale); light-blue band: 95% range if the model were correct
- Raw Chronos-2: too many breaches on the S&P 500 (1.67%, Kupiec $p = 0.001$) and the BET (1.50%)
- Chronos-Bolt, lowest quantile 10% used as "VaR 1%": 13.5%
- Hybrids: breach rates close to 1% in all three markets (average coverage only; full table in the

► [Appendix](#)

The COVID-19 Crash: VaR 1% of Three Models



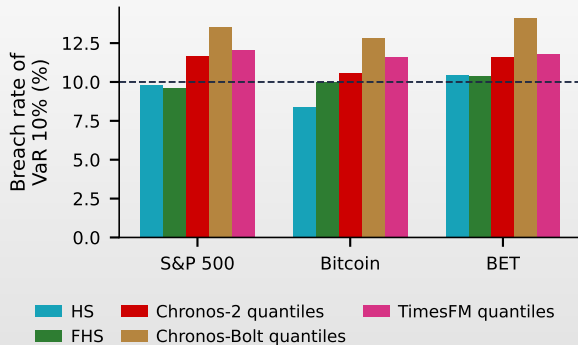
- S&P 500, February–June 2020 (104 days): daily losses and VaR 1% forecasts; dots: breaches
- Breaches: FHS 2, raw Chronos-2 4, FHS with Chronos-2 volatility 2, HS 12; all models are hit in the first days
- Chronos-2 reads the context fast, so its quantiles widen quickly
- The hybrid overshoots: VaR 1% of 58.9% on 17.03.2020, against 26.4% for FHS and a largest loss of 12.8% ([► Appendix](#))

Backtesting Beyond Kupiec: Clustering and the DQ Test

p : indep. / DQ	HS	GARCH-t	FHS	Chronos-2 raw	FHS + Chronos-2	FHS + TimesFM
S&P 500	0.001 / <0.001	0.205 / 0.001	0.045 / 0.021	0.223 / <0.001	0.021 / 0.001	0.029 / <0.001
Bitcoin	0.611 / 0.760	0.261 / 0.463	0.220 / 0.092	0.404 / 0.034	0.026 / 0.004	0.261 / 0.182
BET	0.399 / <0.001	0.258 / 0.082	0.409 / 0.650	0.270 / 0.026	0.475 / <0.001	0.379 / <0.001

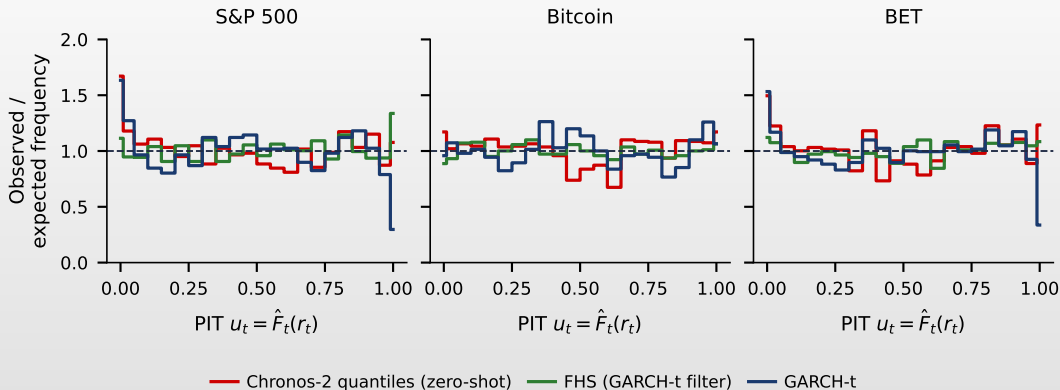
- Kupiec: likelihood-ratio (LR) test that the breach rate equals α ; independence ([Christoffersen \(1998\)](#)): LR test that a breach today does not change the chance of one tomorrow (first-order Markov chain), $\sim \chi^2_1$; conditional coverage = Kupiec + independence, χ^2_2
- Dynamic quantile (DQ, [Engle & Manganelli \(2004\)](#)): regress $I_t - \alpha$ ($I_t = 1$ on a breach day) on a constant, four lagged hits and VaR_t ; Wald statistic that all six coefficients are zero, $\sim \chi^2_6$
- S&P 500: the hybrids have the right breach rate but fail independence and DQ; every model fails DQ at 5%
 - BET: FHS passes both tests, the hybrids fail DQ; right on average is not right conditionally

Where Deciles Suffice: VaR 10%



- Breach rate of the 10% quantile (target 10%) for HS, FHS and the three foundation models
- S&P 500: HS 9.8%, FHS 9.6%; Chronos-2 11.7%, Chronos-Bolt 13.5%, TimesFM-2.5 12.0%
- Kupiec rejects the foundation models in eight of nine cases: the lower tail sits too high, through location, scale or tail shape
- Plausible reason: pretraining corpora are dominated by smoother series; as volatility filters in FHS the models inherit the tails from the data

Calibration of the Whole Distribution: the PIT (1/3)



Calibration of the Whole Distribution: the PIT (2/3)

- PIT (probability integral transform) $u_t = \hat{F}_t(r_t)$: the forecast probability of a return at or below the one realised (Diebold, Gunther & Tay (1998))
 - ▶ $\hat{F}_t$: the predictive distribution function for day t ; if it is correct, the u_t are i.i.d. uniform on $(0, 1)$
 - ▶ bars: observed over expected frequency in the 22 intervals of the Chronos-2 grid; a bar above 1: too many outcomes in that region

 MFM_ch14_calibration

Calibration of the Whole Distribution: the PIT (3/3)

- Berkowitz test ([Berkowitz \(2001\)](#)): transform $z_t = \Phi^{-1}(u_t)$, standard Normal if the forecast is correct; test $\mu = 0$, $\sigma = 1$ for z_t , censored outside [1%, 99%]
 - ▶ Φ^{-1} : standard Normal quantile function; μ , σ : mean and standard deviation of z_t ; $\hat{\sigma} > 1$: tails too thin in the forecast
 - ▶ Chronos-2 rejected on the S&P 500 ($p = 0.019$, $\hat{\sigma} = 1.03$) and the BET; FHS never
- Quantile score: the average pinball loss over the K levels u_k of the grid, a grid analogue of the CRPS (continuous ranked probability score, [Gneiting & Raftery \(2007\)](#))

$$\text{QS} = \frac{2}{K} \sum_{k=1}^K (1\{r_t < q_{u_k}\} - u_k) (q_{u_k} - r_t)$$

- ▶ q_{u_k} : forecast quantile at level u_k ; lower is better
- ▶ Chronos-2 worse than FHS in the centre and the tails (S&P 500 DM +2.51, +3.65); table in the [Appendix](#)

Comparing (VaR 2.5%, ES 2.5%) Forecasts: the MCS

HS	0.140	0.343	0.015
GARCH-t	0.187	0.597	0.341
FHS	1.000	0.597	1.000
Chronos-2 quantiles	0.140	0.641	0.239
FHS + Chronos-2 volatility	0.140	1.000	0.341
FHS + TimesFM volatility	0.183	0.704	0.202
	S&P 500	Bitcoin	BET

- p -values of the 90% model confidence set (Hansen, Lunde & Nason (2011)) for the FZ0 loss; green: kept, red: eliminated; FZ0 table in the [Appendix](#)
- S&P 500: FHS has the lowest FZ0; the hybrids have the right breach rate but clustered breaches and a larger VaR (3.20% vs 2.78%)
- The MCS eliminates only HS on the BET; pairwise DM is sharper: FHS beats the Chronos-2 hybrid on the S&P 500 (DM +2.65, $p = 0.008$) and the BET (DM +2.36)

After the Release: Few Breaches, Little Power

Breaches after 3.11.2025	HS	GARCH-t	FHS	Chronos-2 quantiles	FHS + Chronos-2 vol.	FHS + TimesFM vol.
S&P 500 (220 days)	0	2	2	2	2	2
Bitcoin (320 days)	3	3	4	4	3	4
BET (217 days)	4	4	4	5	1	2

- Expected breaches: 2.2 (S&P 500), 3.2 (Bitcoin), 2.2 (BET): no contamination is possible, but the tests have little power
 - ▶ Exact two-sided Kupiec test at 5% against a true rate of 1.67%: power (probability of rejecting) 0.19 at $T = 220$ days, actual size (rejection rate when the model is right) 0.13; 0.50 at $T = 1,000$; 0.87 at $T = 2,693$
 - ▶ 80% power first at about 2 130 days (8.4 years), for good from 2 320; against 2%: power 0.29 at $T = 220$ (Seminar A8–A9)
- Low power is not zero power: HS has 0 breaches on the S&P 500, and $x = 0$ is in the Kupiec rejection region at $T = 220$; a non-rejection does not establish calibration

Recalibrating Quantiles: Adaptive Conformal Inference (1/2)

- Idea: shift the model's quantile by the error it made recently, and adapt the level after each breach
- Score $E_t = \hat{q}_{\alpha,t} - r_t$; corrected quantile $\hat{q}_{\alpha,t} - \hat{Q}_{1-\alpha_t}$
 - ▶ $\hat{q}_{\alpha,t}$: the model's α -quantile for day t ; $E_t > 0$: the return fell below it
 - ▶ $\hat{Q}_{1-\alpha_t}$: the empirical $(1 - \alpha_t)$ quantile of the last 250 scores
- Adaptive level (Gibbs & Candès (2021))

$$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t), \quad \text{err}_t = 1\{r_t < \hat{q}_{\alpha,t} - \hat{Q}_{1-\alpha_t}\}, \quad \alpha_1 = \alpha$$

- ▶ α : target level (1%); $\text{err}_t = 1$ on a breach; $\gamma > 0$: step size, here 0.001; $\gamma = 0$: plain split conformal
- ▶ After a breach α_t falls (the next VaR is wider); after a calm day it rises slowly

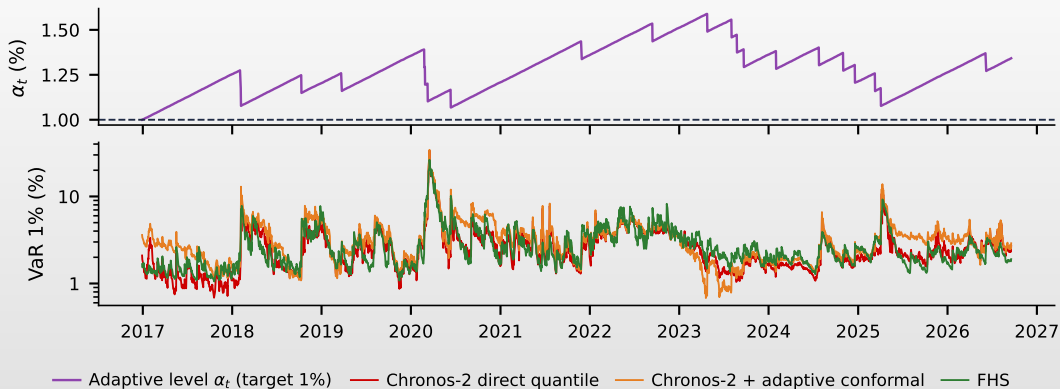
Recalibrating Quantiles: Adaptive Conformal Inference (2/2)

- Guarantee for any sequence of returns, no model needed: over T days the breach rate approaches α

$$\left| \frac{1}{T} \sum_{t=1}^T \text{err}_t - \alpha \right| \leq \frac{\max(\alpha_1, 1 - \alpha_1) + \gamma}{\gamma T}$$

- ▶ the bound shrinks like $1/T$: a long-run statement only, loose for small γ
- ES 2.5%: Chronos-2 ES shifted by the VaR 2.5% correction, a heuristic without a conformal guarantee; rules and the full table in the [Appendix](#)

Adaptive Conformal VaR 1% on the S&P 500 (1/2)



Adaptive Conformal VaR 1% on the S&P 500 (2/2)

- Top: adaptive level α_t (target 1%); bottom: VaR 1% of raw Chronos-2, Chronos-2 with ACI (adaptive conformal inference), FHS (log scale)
- ACI brings the breaches near 1% (S&P 500: 21 vs 24.4 expected); conditional coverage is not rejected in any market
- DQ still rejects on the S&P 500 and the BET; the VaR is wider (3.36% vs 2.85% for FHS) and FZ0 significantly higher
- Proper scoring rules reward sharpness subject to calibration ([Gneiting & Raftery \(2007\)](#))
 - ▶ Calibration is necessary, not sufficient

 MFM_ch14_calibration

Case Study: Is Complexity a Virtue in Return Prediction? (1/2)

□ Questions (Kelly, Malamud & Zhou (2024))

- ▶ must a return-prediction model have far fewer parameters P than observations T ?
- ▶ can models with complexity $c = P/T > 1$ do better out of sample?

□ Setting (eqs. 1–2): next month's return is an unknown function of today's predictors, approximated by a wide network

$$R_{t+1} = f(G_t) + \varepsilon_{t+1}, \quad f(G_t) \approx \sum_{i=1}^P S_{i,t} \beta_i$$

- ▶ R_{t+1} : market excess return; G_t : vector of predictors known at t ; ε_{t+1} : unpredictable part
- ▶ $S_{i,t}$: the i -th non-linear feature of G_t (a hidden unit of a one-hidden-layer network); β_i : its weight; universal approximation (Hornik, Stinchcombe & White (1989))

□ Why it matters: a theoretical case for the large models of this chapter, against the parsimony rule of classical econometrics

Case Study: Is Complexity a Virtue in Return Prediction? (2/2)

- **Link to Test 1:** on the S&P 500 and Bitcoin no model beats the zero forecast significantly; the paper asks whether a negative out-of-sample R^2 means no economic value
 - ▶ Economic value: Sharpe ratio of the market-timing return $\pi_t R_{t+1}$, with position $\pi_t = \hat{\beta}' S_t$ (eq. 6): the forecast itself is the position
 - ▶ $\hat{\beta}$: estimated weights; S_t : vector of the P features at t
- **Theory** (Sections III–IV): under the assumptions of the paper, a misspecified model with optimally chosen ridge shrinkage has expected out-of-sample R^2 and timing Sharpe ratio rising with c
 - ▶ Near $c = 1$ (the interpolation boundary) least squares fits the training data exactly and its variance explodes; beyond it, performance recovers

Case Study: Data and Method (1/2)

- **Data** (Section V.A): monthly excess return of the CRSP value-weighted index; the 15 predictors of [Welch & Goyal \(2008\)](#) (14 variables and the lagged market return), 1926–2020
 - ▶ Returns scaled by their trailing 12-month volatility, predictors by an expanding-window standard deviation; 36 months of warm-up, so the sample starts in 1930
- **Random Fourier features** (eq. 20, [Rahimi & Recht \(2007\)](#)): each feature is a sine or cosine of a random combination of the predictors

$$S_{i,t} = (\sin(\gamma \omega_i' G_t), \cos(\gamma \omega_i' G_t))', \quad \omega_i \sim N(0, I), \quad \gamma = 2$$

- ▶ ω_i : random weight vector, drawn once from the standard Normal distribution (I : identity matrix) and then fixed
- ▶ γ : degree of non-linearity, larger γ gives faster oscillating features
- ▶ A wide two-layer network whose first-layer weights are random; only the output weights β are estimated

Case Study: Data and Method (2/2)

- ▣ **Ridge** (Section II.A), no intercept

$$\hat{\beta}(z) = \left(zI + \frac{1}{T} \sum_t S_t S_t' \right)^{-1} \frac{1}{T} \sum_t S_t R_{t+1}$$

- ▶ $z > 0$: ridge penalty, shrinks $\hat{\beta}$ towards zero; $z \rightarrow 0^+$: the ridgeless (pseudo-inverse) solution; T : window length
- ▣ **Design** (Section V.C): rolling windows of $T = 12, 60, 120$ months; P from 2 to 12 000; $\log_{10} z = -3, \dots, 3$; results averaged over 1 000 independent draws of the features
- ▣ **Evaluation**: out-of-sample $R^2 = 1 - \text{Var}(\text{error})/\text{Var}(R)$ (footnote 40; the theory divides the MSE by $E[R^2]$, the zero forecast of Test 1); annualised Sharpe ratio
 - ▶ alpha and information ratio (alpha over residual volatility) from a regression of the timing return on the market return
 - ▶ Benchmark: the linear "kitchen sink" regression on all 15 predictors (Section V.D)

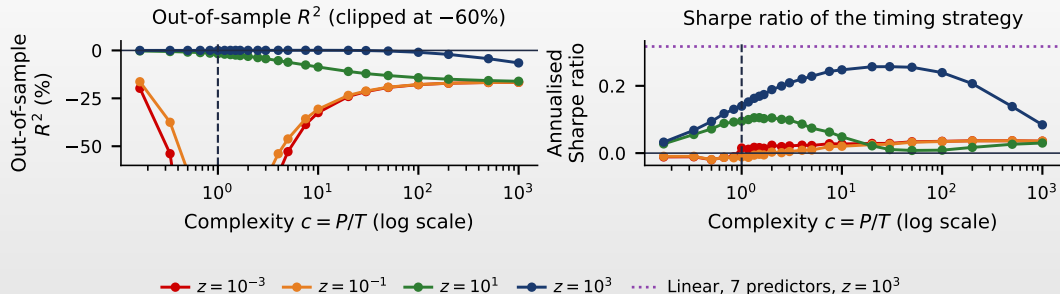
Case Study: Key Results

Model ($T = 12$)	R^2 (%)	Sharpe [t]	Inf. ratio [t]	Max loss	Skewness
Linear, $z = 0^+$	< -100	$-0.11 [-1.0]$	$-0.16 [-1.6]$	98.5	-0.9
Linear, $z = 10^3$	-3.8	$0.46 [4.4]$	$0.33 [3.1]$	2.4	-0.1
RFF, $c = 1000$, $z = 10^3$	0.6	$0.47 [4.5]$	$0.31 [2.9]$	1.2	2.5

Source: Table I, Panel A, of the paper; information ratio against the market; maximum monthly loss in standard deviations

- ▣ Ridgeless kitchen sink ($P = 15 > T = 12$): R^2 below -100% , Sharpe ratio -0.11
 - ▶ Reading: at the interpolation boundary forecasts explode, reproducing the negative verdict of [Welch & Goyal \(2008\)](#)
- ▣ Same predictors with $z = 10^3$: $R^2 = -3.8\%$, yet Sharpe ratio 0.46 ($t = 4.4$)
 - ▶ Reading: R^2 penalises the scale of forecasts; timing profits depend on their correlation with returns (footnote 5)
- ▣ RFF, $c = 1000$: information ratio 0.26 ($t = 2.5$) against the best linear model; with $T = 60$ and 120 , 0.25 and 0.24 against the market ($t = 2.3$ and 2.2 ; Panels B–C)
 - ▶ Reading: nonlinearity extracts more from the same predictors, with a smaller maximum loss and positive skewness
- ▣ Positions (Figure 10): almost always long; cut before 14 of 15 NBER recessions, out of sample
 - ▶ Reading: the model learns the sign restriction of [Campbell & Thompson \(2008\)](#) without being given it

Case Study: Complexity on Course Data (1/2)

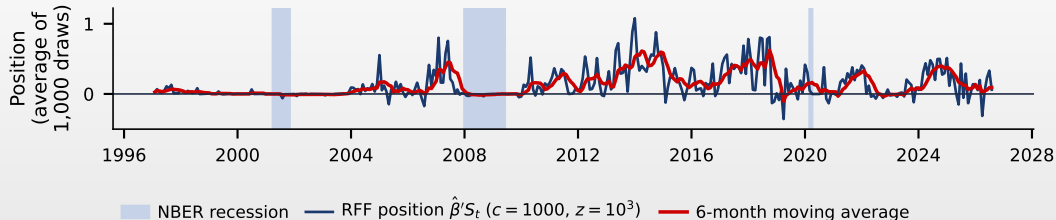


Case Study: Complexity on Course Data (2/2)

- Figures 7–8 of Kelly, Malamud & Zhou (2024) on course data: SPY excess returns, 7 predictors, 354 test months (February 1997–August 2026); $T = 12$, $\gamma = 2$, 1 000 draws; dashed: $c = 1$; dotted: linear, $z = 10^3$
- Nearly ridgeless ($z = 10^{-3}$): $R^2 = -1\,842\%$ at $c = 1$, recovering to -16.6% at $c = 1\,000$: the double descent of Figure 7 (left panel cut at -60%)
- $z = 10^3$: Sharpe ratio 0.14 at $c = 1$, 0.26 at $c = 30$, 0.08 at $c = 1\,000$ ($t = 0.5$); linear with $z = 10^3$: Sharpe 0.32 ($t = 1.7$), not beaten by RFF (information ratio -0.17)
- Reading: complexity helps up to $c \approx 30$, not up to $c = 1\,000$, with 7 of the 15 predictors and 29.5 years

 MFM_ch14_virtue_complexity

Case Study: Timing Positions and Recessions



- Figure 10 of [Kelly, Malamud & Zhou \(2024\)](#) on course data: position $\hat{\beta}' S_t$ of RFF with $c = 1000, z = 10^3$; shaded: NBER recessions (FRED USREC); red: 6-month moving average
- Long in 72% of months, average position 0.13: a positive equity premium learned without a sign restriction
- Average position 0.07 in the 6 months before the 3 recessions and -0.01 during them, against 0.15 otherwise; the 2020 pandemic shock is not anticipated
- Three recessions are a description, not a test: the paper has 15 recessions and 90 years

Case Study: Limits and an Open Question

- **Limits** stated by the authors
 - ▶ Theory for linear ridge models on fixed random features and a single asset; a panel with cross-asset covariances is left open (Introduction, footnote 2)
 - ▶ Predictability spread evenly across features (Assumption 4); sparse LASSO-type models left to future research (footnote 22)
 - ▶ No optimal degree of nonlinearity γ ; average timing returns about half as large in 1975–2020 as in 1930–1974 (Section V.F)
- **Course data** (Seminar C3): 7 of the 15 predictors and about 30 years; the power of a two-sided 5% test for an information ratio of 0.31 is only about 39% ($t \approx 0.31\sqrt{29.5} \approx 1.7$)
- **Open question**: learned instead of random features, the embeddings of a pretrained foundation model such as Chronos-2 as S_t ; is complexity still a virtue, on a panel of course markets?

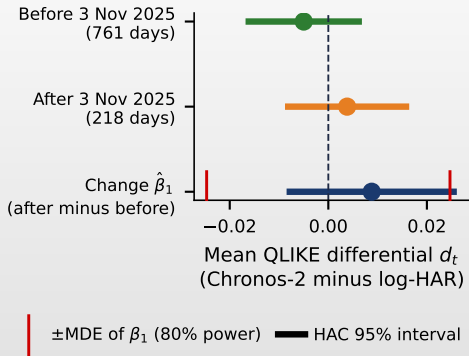
An Open Question: Skill or Memory?

- Test 2: zero-shot Chronos-2 is not distinguishable from log-HAR on SPY realised variance (QLIKE 0.231 vs 0.234, DM $p = 0.537$)
- Rahimikia, Ni & Wang (2025): on daily excess returns across global markets, off-the-shelf foundation models do poorly zero-shot and after fine-tuning; models pretrained from scratch on financial data gain
- Lopez-Lira, Tang & Zhu (2025): once a model has seen the realised values, its counterfactual forecasting skill is not identified; detected memorisation is only a lower bound (shown for LLMs; the logic applies to any pretrained forecaster)
- Open questions
 - ▶ is the parity learned structure (long memory, mean reversion) or recall of the test period?
 - ▶ can a short post-release window tell?
- A test: $d_t = L(\hat{h}_t^{FM}, RV_t) - L(\hat{h}_t^{\log HAR}, RV_t)$, the QLIKE loss L of the foundation model (FM) minus that of log-HAR
 - ▶ memorisation implies $E[d_t | \text{post}] > E[d_t | \text{pre}]$: the FM loses its edge after the release of its weights; H_0 : equality; a rejection may also reflect a regime change

The Discovery Loop in Five Steps

- **Literature:** ask an assistant for evaluations of time-series foundation models on financial data since 2024; check every arXiv identifier and read the corpus documentation yourself
- **Hypotheses:** predictions that separate memorisation from skill: pre vs post release, never-public series, simulated series with the same dynamics
- **Data and code:** from the chapter's volatility forecasts, the QLIKE differential Chronos-2 minus log-HAR, a pre/post test with HAC errors and the MDE; first reproduce the log-HAR QLIKE, 0.234
- **Robustness,** fixed in advance: a placebo release date one year earlier must show no break; simulated series cannot be memorised
- **Critique:** ask the assistant to attack "no memorisation, because the post-release test is insignificant"; expect low power and different regimes (reusable prompts in the [▶ Appendix](#))

Mini-Case: a Test Without Power



- Typical assistant proposal: “the parity of Chronos-2 comes from memorised data, so its edge must vanish after release”
- Mean d_t : -0.0050 before 3 Nov 2025 (761 days), $+0.0038$ after (218 days); change $+0.0088$, HAC $t = 1.00$, $p = 0.32$
- Minimum detectable effect: $MDE = (z_{0.975} + z_{0.8}) se_{HAC}(\hat{\beta}_1) = 0.0247$
 - ▶ $z_{0.975} = 1.96$, $z_{0.8} = 0.84$: standard Normal quantiles, 5% two-sided test, 80% power
 - ▶ $\hat{\beta}_1$: estimated post-release shift in the mean of d_t ; se_{HAC} : its HAC standard error
 - ▶ 4.9 times the pre-release gap; still 0.0169 with an infinite post-release window
- Reading: “no memorisation” is not a finding; power must come from the cross-section, not from waiting

Pitfalls the Researcher Must Check

- **Leakage:** pretraining overlap, full-sample scaling, HAR averages that include day t (Seminar B8 cuts QLIKE by a quarter)
 - ▶ Test: post-release window, simulated or private series; recompute the forecast for t from data truncated at $t - 1$
- **Weak benchmarks:** compare with zero (returns), log-HAR (volatility), FHS (tails), not with a random walk
- **Wrong functional:** medians used as means (mean forecast / mean RV should be near 1; Chronos-2 median: 0.79); deciles used as 1% quantiles
- **Selection:** seeds, contexts and models chosen after seeing the test; log every trial, compare the whole set with the MCS
- **Wrong inference:** DM on nested models, no multiplicity control, coverage without independence, “not significant” reported without the MDE

Your Turn: a Project Seed

- Questions (where memorisation is impossible)
 - ▶ does zero-shot skill for daily variance survive after release?
 - ▶ does it survive on simulated series?
 - ▶ does it survive on less-covered markets?
- Replicate first: the chapter's QLIKE of log-HAR (0.234) and Chronos-2 (0.231), DM $p = 0.537$, and the 218-day post-release window
- Deliverables
 - ▶ A pre-registered power calculation; a panel of series from the course data with a range-based variance proxy and a proxy-robust loss ([Patton \(2011\)](#))
 - ▶ Pre/post tests with placebo release dates; the MCS across models; a referee-style critique and your reply
- Log each prompt that shaped a decision in AI_USE.md; each invented reference, leak or wrong number caught in AI_ERRORS.md, with the test that caught it ([templates](#))

When to Use What

Target	Reference model	Where deep learning can help
Daily index returns	Zero forecast	Less efficient markets (BET); many assets, many features
Realised volatility	log-HAR	Zero-shot forecasts for thousands of series without estimation
VaR 1%, ES 2.5%	FHS (GARCH filter)	Volatility filter in FHS or conformal recalibration

- Foundation models are cheap to try: no estimation, one line of code, a CPU is enough
- Their value is breadth (many series, no tuning), not accuracy on a single well-studied series

Key Takeaways

- ▣ LSTM gates can mitigate vanishing gradients (with a forget-gate bias of 0 they still vanish); attention connects distant time steps directly
- ▣ Foundation models forecast zero-shot from a context window and return quantiles
- ▣ Returns: nothing beats zero on large markets; volatility: foundation models not distinguishable from log-HAR; tails: use them only as volatility filters
- ▣ Evaluate every new model against a strong benchmark and after the release of its weights, with inference that matches the design: Clark–West for nested models, SPA or Romano–Wolf for many models, PIT, quantile scores and DQ for distributions

Check Yourself, and Next: Chapter 15

Retrieval questions

- Why does a stable RNN forget at the rate at which a VAR(1) impulse response decays?
- Why is $\exp(q_{0.5})$ of log RV a biased forecast under QLIKE?
- Why can a hybrid VaR pass Kupiec and still lose the FZ0 comparison?
- Why is “not significant after release” not evidence against memorisation?

Next: LLMs and sentiment analysis

- Can text read by a large language model forecast returns or volatility?
- The same threats return: look-ahead, memorisation of the test period, weak benchmarks
- The same tools: post-release windows, Clark–West, multiplicity control

Appendix: Norms, Spectral Radius and Explosion

- $\gamma_\varphi \|W\|_2 < 1$ is sufficient for geometric vanishing, so $\gamma_\varphi \|W\|_2 \geq 1$ is necessary for explosion (Pascanu, Mikolov & Bengio (2013)); $\|W\|_2$: spectral norm (largest singular value), γ_φ : largest slope of the activation
- Spectral radius ρ (largest absolute eigenvalue): in a linear RNN, $\rho(W) > 1$ is necessary for exponential explosion ($\rho(W) = 1$ with a Jordan block still gives polynomial growth)
 - ▶ With φ the local Jacobians $D_t W$ matter: $\rho(W) = 0$ yet $\rho(DW) > 1$ is possible
- Stable VAR: impulse responses and gradients decay like $\rho(A)^k$ up to a factor k^{m-1} if A has a Jordan block of size m (a repeated eigenvalue without a full set of eigenvectors)
- DUx_t is an innovation only if the inputs x_t are centred white noise; long-memory inputs would pass through the network

Appendix: Geometric versus Hyperbolic Decay, Fitted

- Constant gates, i.i.d. candidate input g_t , stationary start: $\text{Corr}(c_t, c_{t-k}) = f^k$; ARFIMA with $0 < d < 1/2$: $\rho_k \sim Ck^{2d-1}$
- S&P 500 absolute returns since 2000: ACF 0.289 at lag 1, 0.132 at lag 50
 - ▶ Geometric decay with its own amplitude, $\rho_k = \rho_1 f^{k-1}$ (an AR(1) signal observed with noise), matches both lags with $f = 0.984$: $b_f = 4.1$, half-life 43 days
 - ▶ That f implies 0.027 at lag 150; the data show 0.059, more than twice as much (95% band ± 0.024)
- A constant gate $f < 1$ is a short-memory filter; a long context or explicit long averages, as in HAR (Test 2), only approximate long memory over a finite range of lags

Appendix: Other Foundation Models and Their Corpora

- Chronos-Bolt: quantile levels 10%, 20%, . . . , 90% only; Chronos-2: training corpus of selected Chronos and [GIFT-Eval](#) datasets plus synthetic multivariate series; benchmarks GIFT-Eval ([Aksu et al. \(2024\)](#)) and fev-bench
- TimesFM corpus: Google Trends, Wikipedia page views, synthetic series, M4, electricity, traffic, weather; TimesFM-2.5 returns the mean plus deciles
 - ▶ Release dates of the weights, from the Hugging Face repositories: TimesFM-2.5, updated weight version announced on 15.09.2025, released on 01.10.2025; Chronos-2 announced on 20.10.2025, released on 30.10.2025
- **Moirai** ([Woo et al. \(2024\)](#)): any number of variables, mixture-distribution output; Moirai 2.0 ([Liu et al. \(2025\)](#)); non-commercial licence, not used in our tests
- **Lag-Llama** ([Rasul et al. \(2023\)](#)): decoder-only model with lagged values as features and a Student- t output

Appendix: A Formal Contamination Protocol

- Pre/post regression on the loss differential $d_t = L_t^{FM} - L_t^{bench}$: $d_t = \beta_0 + \beta_1 \mathbf{1}\{t \geq \tau\} + u_t$, HAC s.e.; L_t^{FM} , L_t^{bench} : losses of the foundation model and of the benchmark on day t
 - ▶ τ : release date of the weights (a common cutoff when several models are compared); memorisation predicts $\beta_1 > 0$; H_0 : $\beta_1 = 0$
 - ▶ Report the minimum detectable effect (MDE) of β_1 : with 218 post-release days it is 4.9 times the pre-release gap (AI section)
- Pre-registration: forecasts submitted before the outcome exists make contamination impossible by design
 - ▶ Live benchmarks: TS-Arena ([Meyer et al. \(2025\)](#)), Impermanent ([Garza et al. \(2026\)](#)); our post-release window is a manual version
- Ablations: removing the language-model part of LLM-based forecasters does not degrade accuracy, often improves it ([Tan et al. \(2024\)](#))
- Electricity prices: [Pan & Ezzat \(2026\)](#) benchmark on two datasets to limit contamination; foundation models are competitive but do not consistently beat domain-specific methods

Appendix: Return Forecasts, Full Table

R^2 (%) [DM p]	Historical mean	AR(1)	LSTM (5 seeds)	Chronos-2	Chronos-Bolt	TimesFM-2.5
S&P 500	+0.15 [0.095]	+0.94 [0.639]	+0.93 [0.090]	−0.26 [0.418]	−1.57 [0.078]	−3.35 [0.001]
Bitcoin	+0.07 [0.691]	−0.28 [0.326]	+0.04 [0.243]	−0.35 [0.413]	−0.61 [0.070]	−2.29 [0.001]
BET	+0.30 [0.081]	−0.00 [1.000]	+1.15 [0.014]	−0.39 [0.562]	+0.12 [0.831]	−3.27 [0.070]

- Squared error targets the mean; grid means ($\int_0^1 q_u du$) do worse still, e.g. S&P 500 R^2 −0.77% (Chronos-2), −3.42% (TimesFM-2.5)
- DM: two-sided test on squared errors against the zero forecast, Newey–West variance

Appendix: Nested Forecasts, the BET AR(1) and Rolling Windows

- BET AR(1), $R^2 = -0.00\%$ and DM $p = 1.000$, is not a degenerate forecast: s.d. 0.074%, average slope 0.046
 - ▶ Squared-error gains of -12.7 in 2020 and $+14.7$ in 2026 cancel; without these two years $R^2 = -0.11\%$
- Rolling or re-trained models: [Giacomini & White \(2006\)](#) test the forecasting method, estimation window included ([Diebold \(2015\)](#) on the use and abuse of DM)
- SPA refines White's Reality Check ([White \(2000\)](#)): very poor models (TimesFM) no longer dilute the test; both use a stationary bootstrap with mean block 20 days

Appendix: QLIKE, DM and MCS, Full Table

Model	QLIKE	DM vs log-HAR	p	MCS p	QLIKE after 3.11.2025
Random walk	0.326	+4.53	<0.001	0.00	0.384
HAR	0.247	+1.30	0.195	0.29	0.270
log-HAR	0.234	—	—	0.95	0.238
GARCH-t	0.283	+3.43	<0.001	0.03	0.295
LSTM	0.284	+1.68	0.094	0.29	0.233
Chronos-2	0.231	−0.62	0.537	1.00	0.242
Chronos-Bolt	0.232	−0.18	0.857	0.95	0.236
TimesFM-2.5	0.233	−0.08	0.938	0.95	0.244

- Literature: [Brini \(2026\)](#) compares nine zero-shot foundation models with eight HAR-type models on 50 assets; pooled losses favour foundation models because of a few outlier assets, while asset by asset only one small model beats log-HAR at every horizon, narrowly
- [Goel et al. \(2025\)](#): TimesFM zero-shot is a reasonable baseline; incremental fine-tuning is needed to beat econometric models

Appendix: VaR 1% Breaches and Kupiec Tests, Full Table

Breaches [Kupiec p]	HS	GARCH-t	FHS	Chronos-2 quantiles	FHS + Chronos-2 vol.	FHS + TimesFM vol.
S&P 500 (26.9)	37 [0.065]	44 [0.002]	30 [0.559]	45 [0.001]	25 [0.705]	27 [0.989]
Bitcoin (28.2)	19 [0.065]	27 [0.822]	25 [0.539]	33 [0.374]	27 [0.822]	27 [0.822]
BET (26.8)	32 [0.322]	41 [0.010]	30 [0.536]	40 [0.016]	26 [0.884]	32 [0.322]

- In brackets after the market: the expected number of breaches, 1% of the days
- Kupiec: two-sided likelihood-ratio test of a 1% breach rate, χ_1^2

Appendix: COVID-19 Crash, Scaling and Overshoot

- Chronos-2 standardises the context by its mean and standard deviation, then applies $\sinh^{-1}$ (Ansari et al. (2025)); on 16.03.2020 the crash raised the 512-day standard deviation by 9.7% (mean absolute scale of Chronos: 3.2%, Seminar A4)
 - ▶ So the widening mixes the scale change and the reading of the crash days; only an ablation could separate them
- Not a unit error: after the crash day Chronos-2 forecast a mean $|r|$ of 12.5% for the next day, then FHS multiplies it by the 1% quantile of $r/\hat{\sigma}$ (4.7)
 - ▶ The filter extrapolates a one-day spike as a new level: VaR 4.6 times the largest loss, a high capital cost that the FZ0 loss penalises

Appendix: Berkowitz Tests and Quantile Scores, Three Markets

Market	Berkowitz p : Chronos-2	FHS	Below $q_{1\%}$ (C2)	QS: C2 / FHS	DM [p]
S&P 500	0.019	0.599	1.67%	0.510 / 0.506	+3.06 [0.002]
Bitcoin	0.595	0.306	1.17%	1.526 / 1.529	-0.90 [0.369]
BET	0.045	0.436	1.50%	0.465 / 0.460	+3.48 [<0.001]

- The Berkowitz test evaluates the linearly interpolated grid distribution function and only μ, σ , not serial independence of the PIT
- QS weights the 21 levels equally; the CRPS would weight each level by its spacing; C2: Chronos-2; tie on Bitcoin

Appendix: FZ0 Losses and MCS p -Values, Full Table

FZ0 [MCS p]	HS	GARCH-t	FHS	Chronos-2 quantiles	FHS + Chronos-2 vol.	FHS + TimesFM vol.
S&P 500	1.309 [0.14]	1.026 [0.19]	0.999 [1.00]	1.147 [0.14]	1.084 [0.14]	1.059 [0.18]
Bitcoin	2.237 [0.34]	2.192 [0.60]	2.192 [0.60]	2.181 [0.64]	2.151 [1.00]	2.160 [0.70]
BET	1.230 [0.01]	1.001 [0.34]	0.963 [1.00]	1.032 [0.24]	1.007 [0.34]	1.027 [0.20]

- Bitcoin: the hybrid FHS with Chronos-2 volatility has the lowest FZ0 loss, but the differences are not significant
- FZ0 for (VaR 2.5%, ES 2.5%); the MCS uses the $T_{\max}$ statistic and a block bootstrap

Appendix: Conformal Recalibration, Rules and Results

After 250 calibration days:				
breaches [cc p / DQ p]	FHS	Chronos-2 raw	FHS + C2 vol.	Chronos-2 + ACI
S&P 500 (24.4)	28 [0.097 / 0.014]	42 [0.002 / <0.001]	23 [0.058 / <0.001]	21 [0.307 / 0.049]
Bitcoin (25.7)	24 [0.449 / 0.108]	31 [0.410 / 0.026]	26 [0.087 / 0.004]	24 [0.449 / 0.191]
BET (24.2)	26 [0.709 / 0.727]	36 [0.048 / 0.184]	23 [0.776 / 0.011]	25 [0.762 / 0.032]
FZ0, S&P 500 [DM p]	1.013	1.142 [0.009]	1.087 [0.022]	1.205 [0.001]

- $\hat{Q}$: the $\lceil (1 - \alpha_t)(250 + 1) \rceil$ -th smallest past score ($\lceil \cdot \rceil$: rounding up), $+\infty$ if that rank exceeds 250 or $\alpha_t \leq 0$, $-\infty$ if $\alpha_t \geq 1$; with $\gamma = 0.005$: infinite VaR on 24 (S&P 500), 100 (Bitcoin), 270 (BET) days
- Guarantee with $T = 2443$, $\gamma = 0.001$: bound 0.41, uninformative for a 1% rate; without exchangeability, weighted conformal bounds the gap by the drift ([Barber et al. \(2023\)](#))
- ES comparisons are harder still: finite-sample precision limits ([Pele & Mazurencu-Marinescu-Pele \(2026\)](#)); ES is backtestable jointly with VaR ([Nolde & Ziegel \(2017\)](#)); a GPD tail ([McNeil & Frey \(2000\)](#)) is an alternative for ES

Appendix: Case Study, Replication with Course Data

- Monthly SPY total returns from 1993 (course data) in excess of the 3-month Treasury bill rate (FRED TB3MS)
 - ▶ Predictors of [Welch & Goyal \(2008\)](#) available: Treasury-bill rate (tb1), long-term yield (lty), term spread (tms), default yield spread (dfy), inflation (infl), all from FRED; stock variance (svar) from daily S&P 500 returns; the lagged excess return: 7 of the 15
 - ▶ Data limitation: the valuation ratios (dividend-price dp, dividend yield dy, earnings-price ep, payout de, book-to-market b/m), net equity expansion (ntis) and the bond returns (long-term return ltr, default return spread dfr) are not in the course data; about 30 years instead of 90
 - ▶ Method unchanged: $\gamma = 2$, $T = 12$, $P \leq 12\,000$, $\log_{10} z = -3, \dots, 3$, 1 000 draws
- Course-data results: linear kitchen sink, $z = 0^+$ ($c = 7/12$): $R^2 = -1\,540\%$, Sharpe -0.06 ; linear, $z = 10^3$: $R^2 = 0.26\%$, information ratio 0.25 ($t = 1.3$) against SPY; RFF, $c = 1\,000$: $R^2 = -6.5\%$, information ratio -0.17 ($t = -0.9$)

Appendix: The Discovery Loop, Reusable Prompts (1/2)

- **Literature:** "Find evaluations of time-series foundation models on financial returns or volatility since 2024: models, data, loss, benchmark, and whether the test period overlaps the pretraining corpus."
 - ▶ Semantic Scholar, Google Scholar, Elicit help with the search
- **Hypotheses:** "List predictions that separate memorisation from skill for a zero-shot volatility forecaster: pre vs post release, never-public series, simulated series with the same dynamics."
 - ▶ Diagnostics, not proof: a simulated log-HAR path tests transfer to that process; a pre/post gap can also reflect a regime change
- **Data and code:** "From the daily SPY realised-variance forecasts of Chronos-2 and log-HAR, compute the QLIKE differential Chronos-2 minus log-HAR, test pre vs post 3 November 2025 with HAC errors, and report the minimum detectable effect at 80% power."
 - ▶ Any assistant (Claude, ChatGPT, Gemini, GitHub Copilot)

Appendix: The Discovery Loop, Reusable Prompts (2/2)

- **Robustness:** "Design a pre-registered test with more power: which cross-section of series, which variance proxy, which panel DM test, and how many post-release days."
- **Critique:** "Attack this claim: no memorisation, because the post-release DM test is insignificant. What is wrong with the inference?"
 - ▶ Expect: low power, absence of evidence, and different regimes: the April 2025 tariff shock lies in the pre-release sample, the post-release window is calmer
- **Report:** model versions and weight release dates, context length, mean or median, every seed and window tried

Appendix: Further Reading

- Gu, Kelly & Xiu (2020): machine learning for the cross-section of expected returns
- Kelly, Malamud & Zhou (2024): why very large models can predict returns better
- Ansari et al. (2024): Chronos, time series read as a language
- Crypto applications: liquidity forecasting Agakishiev, Härdle & Pele (2025), early warning for “zombie” crypto-assets Będowska-Sójka, Wójcik & Pele (2026), statistical classification of crypto-assets Pele et al. (2023)
- Textbooks: Franke, Härdle & Hafner (2019) and Härdle, Chen & Overbeck (2017) for the statistics of financial markets behind the benchmarks

References (1/7)

- ▣ Agakishiev, I., Härdle, W.K., Pele, D.T. (2025). [Cryptocurrency liquidity forecasting](#). In *Springer Handbooks of Computational Statistics*, 41–60. Springer.
- ▣ Aksu, T., Woo, G., Liu, J., Liu, X., Liu, C., Savarese, S., Xiong, C., Sahoo, D. (2024). [GIFT-Eval: a benchmark for general time series forecasting model evaluation](#). arXiv:2410.10393.
- ▣ Andersen, T.G., Bollerslev, T., Diebold, F.X., Labys, P. (2003). [Modeling and forecasting realized volatility](#). *Econometrica*, 71(2), 579–625.
- ▣ Ansari, A.F., Shchur, O., Küken, J., Auer, A., et al. (2025). [Chronos-2: from univariate to universal forecasting](#). arXiv:2510.15821.
- ▣ Ansari, A.F., Stella, L., Turkmen, C., Zhang, X., et al. (2024). [Chronos: learning the language of time series](#). *Transactions on Machine Learning Research*; arXiv:2403.07815.
- ▣ Barber, R.F., Candès, E.J., Ramdas, A., Tibshirani, R.J. (2023). [Conformal prediction beyond exchangeability](#). *Annals of Statistics*, 51(2), 816–845.
- ▣ Będowska-Sójka, B., Wójcik, P., Pele, D.T. (2026). [Early warning systems for cryptocurrency markets: predicting 'zombie' assets using machine learning](#). *North American Journal of Economics and Finance*, 81, 102543.
- ▣ Bengio, Y., Simard, P., Frasconi, P. (1994). [Learning long-term dependencies with gradient descent is difficult](#). *IEEE Transactions on Neural Networks*, 5(2), 157–166.
- ▣ Berkowitz, J. (2001). [Testing density forecasts, with applications to risk management](#). *Journal of Business & Economic Statistics*, 19(4), 465–474.

References (2/7)

- ▣ Brini, A. (2026). [Forecasting realized volatility with time series foundation models: a comparison with econometric benchmarks.](#) [arXiv:2607.05291](#).
- ▣ Campbell, J.Y., Thompson, S.B. (2008). [Predicting excess stock returns out of sample: can anything beat the historical average?](#) *Review of Financial Studies*, 21(4), 1509–1531.
- ▣ Christoffersen, P.F. (1998). [Evaluating interval forecasts.](#) *International Economic Review*, 39(4), 841–862.
- ▣ Clark, T.E., West, K.D. (2007). [Approximately normal tests for equal predictive accuracy in nested models.](#) *Journal of Econometrics*, 138(1), 291–311.
- ▣ Corsi, F. (2009). [A simple approximate long-memory model of realized volatility.](#) *Journal of Financial Econometrics*, 7(2), 174–196.
- ▣ Das, A., Kong, W., Sen, R., Zhou, Y. (2024). [A decoder-only foundation model for time-series forecasting.](#) *Proceedings of ICML*, PMLR 235; [arXiv:2310.10688](#).
- ▣ Diebold, F.X. (2015). [Comparing predictive accuracy, twenty years later: a personal perspective on the use and abuse of Diebold–Mariano tests.](#) *Journal of Business & Economic Statistics*, 33(1), 1.
- ▣ Diebold, F.X., Gunther, T.A., Tay, A.S. (1998). [Evaluating density forecasts with applications to financial risk management.](#) *International Economic Review*, 39(4), 863–883.
- ▣ Diebold, F.X., Mariano, R.S. (1995). [Comparing predictive accuracy.](#) *Journal of Business & Economic Statistics*, 13(3), 253–263.

References (3/7)

- ▣ Engle, R.F., Manganelli, S. (2004). [CAViaR: conditional autoregressive value at risk by regression quantiles](#). *Journal of Business & Economic Statistics*, 22(4), 367–381.
- ▣ Fischer, T., Krauss, C. (2018). [Deep learning with long short-term memory networks for financial market predictions](#). *European Journal of Operational Research*, 270(2), 654–669.
- ▣ Fissler, T., Ziegel, J.F. (2016). [Higher order elicibility and Osband's principle](#). *Annals of Statistics*, 44(4), 1680–1707.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). [Statistics of Financial Markets: An Introduction](#), 5th ed. Springer.
- ▣ Garza, A., Rosillo, R., Mendoza-Smith, R., Salinas, D., Williams, A.R., Ashok, A. (2026). [Impermanent: a live benchmark for temporal generalization in time series forecasting](#). [arXiv:2603.08707](#).
- ▣ Giacomini, R., White, H. (2006). [Tests of conditional predictive ability](#). *Econometrica*, 74(6), 1545–1578.
- ▣ Gibbs, I., Candès, E. (2021). [Adaptive conformal inference under distribution shift](#). *NeurIPS*; [arXiv:2106.00170](#).
- ▣ Gneiting, T. (2011). [Making and evaluating point forecasts](#). *Journal of the American Statistical Association*, 106(494), 746–762.
- ▣ Gneiting, T., Raftery, A.E. (2007). [Strictly proper scoring rules, prediction, and estimation](#). *Journal of the American Statistical Association*, 102(477), 359–378.

References (4/7)

- ▣ Goel, A., Pasricha, P., Magris, M., Kannianen, J. (2025). [Foundation time-series AI model for realized volatility forecasting](#). [arXiv:2505.11163](#).
- ▣ Gu, S., Kelly, B., Xiu, D. (2020). [Empirical asset pricing via machine learning](#). *Review of Financial Studies*, 33(5), 2223–2273.
- ▣ Hansen, P.R. (2005). [A test for superior predictive ability](#). *Journal of Business & Economic Statistics*, 23(4), 365–380.
- ▣ Hansen, P.R., Lunde, A., Nason, J.M. (2011). [The model confidence set](#). *Econometrica*, 79(2), 453–497.
- ▣ Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (eds.) (2017). [Applied Quantitative Finance](#), 3rd ed. Springer.
- ▣ Hochreiter, S., Schmidhuber, J. (1997). [Long short-term memory](#). *Neural Computation*, 9(8), 1735–1780.
- ▣ Hornik, K., Stinchcombe, M., White, H. (1989). [Multilayer feedforward networks are universal approximators](#). *Neural Networks*, 2(5), 359–366.
- ▣ Kelly, B., Malamud, S., Zhou, K. (2024). [The virtue of complexity in return prediction](#). *Journal of Finance*, 79(1), 459–503.
- ▣ Kingma, D.P., Ba, J. (2015). [Adam: a method for stochastic optimization](#). *ICLR*; [arXiv:1412.6980](#).

References (5/7)

- ▣ LeCun, Y., Bengio, Y., Hinton, G. (2015). [Deep learning](#). *Nature*, 521(7553), 436–444.
- ▣ Liu, C., Aksu, T., Liu, J., Liu, X., et al. (2025). [Moirai 2.0: when less is more for time series forecasting](#). arXiv:2511.11698.
- ▣ Lo, A.W., MacKinlay, A.C. (1990). [An econometric analysis of nonsynchronous trading](#). *Journal of Econometrics*, 45(1–2), 181–211.
- ▣ Lopez-Lira, A., Tang, Y., Zhu, M. (2025). [The memorization problem: can we trust LLMs' economic forecasts?](#) arXiv:2504.14765.
- ▣ McNeil, A.J., Frey, R. (2000). [Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach](#). *Journal of Empirical Finance*, 7(3–4), 271–300.
- ▣ Meyer, M., Kaltenpoth, S., Albers, H., Zalipski, K., Müller, O. (2025). [TS-Arena: a live forecast pre-registration platform](#). arXiv:2512.20761.
- ▣ Nie, Y., Nguyen, N.H., Sinthong, P., Kalagnanam, J. (2023). [A time series is worth 64 words: long-term forecasting with Transformers](#). *ICLR*; arXiv:2211.14730.
- ▣ Nolde, N., Ziegel, J.F. (2017). [Elicitability and backtesting: perspectives for banking regulation](#). *Annals of Applied Statistics*, 11(4), 1833–1874.
- ▣ Oreshkin, B.N., Carpov, D., Chapados, N., Bengio, Y. (2020). [N-BEATS: neural basis expansion analysis for interpretable time series forecasting](#). *ICLR*; arXiv:1905.10437.

References (6/7)

- ▣ Pan, Z., Ezzat, A.A. (2026). [Evaluating time series foundation models for electricity price forecasting: contamination risk, distributional shifts, and covariate dependence](#). [arXiv:2607.02623](#).
- ▣ Pascanu, R., Mikolov, T., Bengio, Y. (2013). [On the difficulty of training recurrent neural networks](#). *ICML*; [arXiv:1211.5063](#).
- ▣ Patton, A.J. (2011). [Volatility forecast comparison using imperfect volatility proxies](#). *Journal of Econometrics*, 160(1), 246–256.
- ▣ Patton, A.J., Ziegel, J.F., Chen, R. (2019). [Dynamic semiparametric models for expected shortfall \(and value-at-risk\)](#). *Journal of Econometrics*, 211(2), 388–413.
- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2026). [Finite-sample precision limits for expected shortfall forecast comparisons](#). *Mathematics*, 14(13), 2316.
- ▣ Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). [Are cryptos becoming alternative assets?](#) *The European Journal of Finance*, 29(10), 1064–1105.
- ▣ Rahimi, A., Recht, B. (2007). [Random features for large-scale kernel machines](#). *Advances in Neural Information Processing Systems*, 20.
- ▣ Rahimikia, E., Ni, H., Wang, W. (2025). [Re\(Visiting\) time series foundation models in finance](#). [arXiv:2511.18578](#).
- ▣ Rasul, K., Ashok, A., Williams, A.R., Ghonia, H., et al. (2023). [Lag-Llama: towards foundation models for probabilistic time series forecasting](#). [arXiv:2310.08278](#).

References (7/7)

- ▣ Romano, J.P., Wolf, M. (2005). [Stepwise multiple testing as formalized data snooping](#). *Econometrica*, 73(4), 1237–1282.
- ▣ Rumelhart, D.E., Hinton, G.E., Williams, R.J. (1986). [Learning representations by back-propagating errors](#). *Nature*, 323(6088), 533–536.
- ▣ Salinas, D., Flunkert, V., Gasthaus, J., Januschowski, T. (2020). [DeepAR: probabilistic forecasting with autoregressive recurrent networks](#). *International Journal of Forecasting*, 36(3), 1181–1191.
- ▣ Tan, M., Merrill, M.A., Gupta, V., Althoff, T., Hartvigsen, T. (2024). [Are language models actually useful for time series forecasting?](#) *NeurIPS*; arXiv:2406.16964.
- ▣ Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., et al. (2017). [Attention is all you need](#). *NeurIPS*; arXiv:1706.03762.
- ▣ Welch, I., Goyal, A. (2008). [A comprehensive look at the empirical performance of equity premium prediction](#). *Review of Financial Studies*, 21(4), 1455–1508.
- ▣ White, H. (2000). [A reality check for data snooping](#). *Econometrica*, 68(5), 1097–1126.
- ▣ Woo, G., Liu, C., Kumar, A., Xiong, C., Savarese, S., Sahoo, D. (2024). [Unified training of universal time series forecasting Transformers](#). *ICML*; arXiv:2402.02592.